Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering Under Extreme Imbalance and Topological Camouflage

Title Page & Author Declarations (Engineering Applications of Artificial Intelligence)

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CRediT Author Statement

- **Md. Nazmul:** Conceptualization, Methodology, Software Architecture, Mathematical Derivations, Empirical Evaluation, Investigation, Writing – Original Draft, Visualization, Project Administration.
- **Musrat Jahan Gungun:** Data Curation, Validation, Statistical Significance Testing, Baseline Benchmarking, Writing – Review & Editing.
- **Maheli Ahmed:** Supervision, Resources, Funding Acquisition, Formal Analysis, Model Governance Alignment, Writing – Review & Editing.

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Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Submission Declaration & Verification

The authors confirm that this manuscript represents original research that has not been published previously and is not under consideration for publication elsewhere. Publication has been approved by all authors and tacitly or explicitly by the responsible authorities where the work was carried out. If accepted, it will not be published elsewhere in the same form, in English or in any other language, including electronically, without the written consent of the copyright holder.

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Research Data and Code Availability Statement

All 14 financial benchmark datasets evaluated in this study (Elliptic-v1, Elliptic-v2, XBlock-ETH, Mt-Gox, SAML-D, PaySim, PaySim Extended, IBM-AMLSim HI/LI regimes, DGraphFin, and Credit Card Forensics) are derived from openly accessible public repositories. The synthetic typology data generator is provided with the project software. Complete end-to-end pipelines, model checkpoints, environment Docker specifications, and verification test suites are publicly maintained for double-blind peer review at: https://anonymous.4open.science/r/Intelligent-AML. A permanent citable Zenodo DOI and public GitHub repository will be minted upon acceptance.